

Kedarnath Mohan

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EDUCATION

University of Georgia

GPA: 3.9 | Expected May 2028

Bachelor of Science, Major in Computer Engineering, Minor in Astrophysics

Athens, GA

Relevant Coursework: Data Structures & Alg., Sys. Programming, Deep Learning, Computer Vision, Linear Algebra

EXPERIENCE

Amazon | Software Development Intern

May 2026 – Aug. 2026

Kindle Sync | Java, JSP, GraphQL, Tomcat, DynamoDB

Seattle, WA

- Upgraded a Java web app running on Tomcat to give 100+ developers self-service sync debugging.
- Secured 1M+ customer records from data exposure by enforcing namespace-level access control across 15+ domains.
- Removed 100% of debugging API load by adding direct cross-account DynamoDB reads secured with AWS IAM roles.
- Surfaced data integrity issues by building a diff engine to compare live API data with direct DynamoDB reads.

Zhao Nano Lab | Software Engineer Intern

Jan. 2026 – May 2026

University of Georgia | Python, NumPy, SciPy, Pandas

Athens, GA

- Supported 1,500+ global users by maintaining open-source spectra analysis software for Raman spectroscopy.
- Improved SNR by 38.7% vs standard wavelet and 24.9% vs FFT using robust wavelet denoising on Raman spectra.
- Automated data normalization workflows for heterogeneous datasets using Python, NumPy, SciPy, and Pandas.
- Improved edge-case accuracy by benchmarking pipelines and tuning default parameters with telemetry-driven metrics.

HammerHead Construction | Software Engineering Intern

Mar. 2025 – Jan. 2026

University of Georgia | Python, React, TypeScript, YOLOv9

Athens, GA

- Converted 2D architectural blueprints into accurate 3D models by engineering a geometry synthesis engine in Python.
- Reduced annotation time by 83% by fine-tuning a YOLOv9 computer vision pipeline for CAD-ready detection.
- Raised test coverage to 85% by adding Vitest unit tests for React/TypeScript export and validation workflows.

PROJECTS

Real-Time Arbitrage Engine | C++, Erlang/OTP, FastAPI, Redis, PostgreSQL, React

- Achieved sub-20ms ingestion latency for live market feeds by building a concurrent stream processor in Erlang/OTP.
- Maintained strict price-time priority with $O(\log N)$ complexity by implementing a Redis Sorted Set order book.
- Identified risk-free arbitrage opportunities by standardizing market schemas into a unified pricing structure.
- Ensured zero data loss during crashes by externalizing volatile state into Redis with a React dashboard to view logs.

Water Quality Data Monitoring Platform | Python, React, TypeScript, WebSockets, PostgreSQL

- Built a full-stack platform to stream, analyze, and visualize live sensor data from distributed test devices.
- Processed 5M+ sensor samples by building a Python ingestion pipeline with validation and anomaly detection.
- Cut analysis time by 38% by adding synchronized plots, event markers, and automated test-run summaries.
- Supported live monitoring across 20+ sensors by streaming validated measurements to a React dashboard.

LEADERSHIP & INVOLVEMENT

Executive Board Member | AI @ UGA

Aug. 2025 – Present

- Secure industry-sponsored AI projects with firms and faculty to provide technical experience for 100+ members.
- Collaborate with engineers and project managers to communicate milestones across 10+ projects for various firms.
- Orchestrate technical recruitment and interview cycles for 50+ candidates to evaluate proficiency and team caliber.

TECHNICAL SKILLS

Languages: Python, C++, C, Java, SQL, Erlang/OTP, Lua, HTML/CSS

Frameworks & Libraries: React, Node.js, Spring Boot, FastAPI, TF Lite, OpenCV, NumPy, SciPy

Infrastructure & Tools: AWS (S3, EC2, ECS, Lambda), GCP (Pub/Sub, Vertex, GKE), Docker, Kubernetes, Git, GitHub Actions, Redis, PostgreSQL, Supabase, WebSockets, OpenTelemetry